

Proposed End-to-End Pipeline for Physics-Aware Hyperspectral Analysis of Spectrometer Data

The current direction of the project is to develop a physically grounded pipeline for hyperspectral analysis using spectrometer measurements collected in a controlled laboratory setup. The overall workflow combines radiometric correction, background removal, attenuation modeling, and physics-aware hyperspectral unmixing. The objective is to recover physically meaningful bottom reflectance spectra and subsequently decompose those spectra into constituent spectral components using constrained deep learning methods.

The proposed methodology is motivated by shallow-water remote sensing and benthic reflectance retrieval problems commonly studied in ocean optics. However, unlike satellite remote sensing applications, our setup benefits from controlled illumination, short optical path lengths, fixed geometry, and negligible atmospheric effects. These simplifications allow us to directly model and invert the radiative transfer process with fewer assumptions and reduced uncertainty.

The full processing pipeline currently consists of the following stages:

1. Raw spectral acquisition
2. Spectral preprocessing and quality control
3. Outlier detection and removal
4. Water/background correction
5. Attenuation estimation and reflectance inversion
6. Physics-aware hyperspectral unmixing using constrained autoencoders

The first stage involves acquiring hyperspectral measurements using the spectrometer system under controlled illumination conditions. The measured spectra contain contributions from the target material, water background, attenuation effects, sensor noise, and experimental variability. Consequently, preprocessing and correction become essential before any machine learning analysis is performed.

The preprocessing stage includes dark-current correction, white-reference normalization, and optional spectral smoothing. Dark-current correction removes the intrinsic electronic response of the sensor when no light is present. White-reference normalization converts the measurements into relative reflectance by dividing by a calibrated white standard spectrum. Spectral smoothing may additionally be applied to suppress high-frequency noise while preserving the overall spectral structure.

After preprocessing, the next step is outlier detection and removal. This stage is important because hyperspectral measurements are highly sensitive to illumination fluctuations, air bubbles, sensor instability, scattering artifacts, and experimental perturbations. Spectral outliers can significantly bias attenuation estimation and degrade downstream unmixing performance. The outlier removal process therefore aims to retain only physically consistent spectra prior to inversion and modeling.

Following outlier removal, the main physical correction stage is performed. The measured reflectance is modeled using a simplified radiative transfer formulation adapted from shallow-water optics:

$$R_m(\lambda) = R_w(\lambda) + (R_b(\lambda) - R_w(\lambda))e^{-2K_d(\lambda)z}$$

where:

- $R_m(\lambda)$ is the measured reflectance,
- $R_w(\lambda)$ is the water/background reflectance,
- $R_b(\lambda)$ is the true bottom reflectance,
- $K_d(\lambda)$ is the wavelength-dependent attenuation coefficient,
- and z is the water path length.

The factor of 2 in the exponential term represents the two-way attenuation of light through the water column. Light is attenuated once along the downward path from the illumination source to the target and again along the upward path from the target back to the spectrometer sensor.

The exponential attenuation model originates from the Beer–Lambert law. The negative exponent reflects the physical decay of light intensity due to absorption and scattering processes within the medium. Importantly, this model satisfies physically meaningful boundary conditions. At zero depth ($z = 0$), the attenuation term becomes unity and the measured reflectance equals the true bottom reflectance:

$$R_m = R_b$$

At sufficiently large depths, the attenuation term approaches zero and the measured signal converges toward the water/background reflectance:

$$R_m \rightarrow R_w$$

This behavior confirms the physical consistency of the formulation.

The next stage consists of estimating attenuation effects and recovering the bottom reflectance spectra. The preferred approach is to estimate wavelength-dependent attenuation coefficients using controlled multi-depth measurements. Taking the logarithm of the corrected signal yields:

$$\ln (R_m - R_w) = \ln (R_b - R_w) - 2K_d z$$

This expression linearizes the attenuation process and allows attenuation coefficients to be estimated through linear regression across different path lengths. If exact depth measurements are unavailable, the physically identifiable quantity becomes the optical thickness:

$$\tau(\lambda) = 2K_d(\lambda)z$$

which represents the total attenuation accumulated along the two-way optical path. In this case, attenuation can still be estimated directly without explicitly separating K_d and z .

Once attenuation has been estimated, the corrected bottom reflectance can be recovered through direct inversion of the physical model:

$$R_b(\lambda) = R_w(\lambda) + (R_m(\lambda) - R_w(\lambda))e^{2K_d(\lambda)z}$$

The resulting spectra represent the estimated intrinsic reflectance of the bottom material with water effects minimized or removed.

After radiometric correction and reflectance retrieval, the final stage of the project focuses on hyperspectral unmixing using a physics-aware autoencoder framework. Traditional autoencoders learn latent representations purely from data and generally ignore the physical constraints governing hyperspectral mixing processes. In contrast, the proposed framework incorporates constraints derived from the linear mixing model commonly used in hyperspectral imaging.

The corrected spectrum is modeled as:

$$x(\lambda) \approx \sum_{i=1}^p a_i s_i(\lambda)$$

where:

- $s_i(\lambda)$ are endmember spectra,
- a_i are abundance fractions,
- and p is the number of spectral components.

The abundances are constrained such that:

$$a_i \geq 0$$

and:

$$\sum_i a_i = 1$$

These constraints enforce physical interpretability and ensure that abundances represent meaningful fractional contributions.

The proposed neural architecture consists of an encoder and a physically interpretable decoder. The encoder maps each input spectrum into abundance vectors, while the decoder reconstructs the spectrum as a weighted linear combination of learned endmembers. Unlike generic nonlinear autoencoders, the decoder is intentionally designed to remain linear:

$$\hat{x} = Ea$$

where E is the learned endmember matrix and a represents abundance vectors. This formulation preserves interpretability because the decoder weights directly correspond to physically meaningful spectral signatures.

Additional physics-aware constraints can also be incorporated into the training objective, including sparsity regularization, abundance sum-to-one penalties, nonnegativity constraints, and spectral smoothness regularization. These additions help stabilize the learned representations and suppress physically unrealistic solutions.

A more advanced extension of the framework may later integrate attenuation modeling directly into the decoder itself. In that case, the reconstruction process would jointly model abundance mixing and water attenuation:

$$x = R_w + (Ea - R_w)e^{-\tau}$$

Such a formulation would allow the network to simultaneously learn endmembers, abundances, and attenuation effects within a unified physics-informed framework.

Overall, the proposed pipeline combines physically grounded radiative transfer modeling with modern representation learning techniques. By performing correction prior to unmixing and embedding physical constraints within the neural network architecture, the framework aims to produce more interpretable, robust, and scientifically meaningful hyperspectral analyses than purely data-driven approaches.